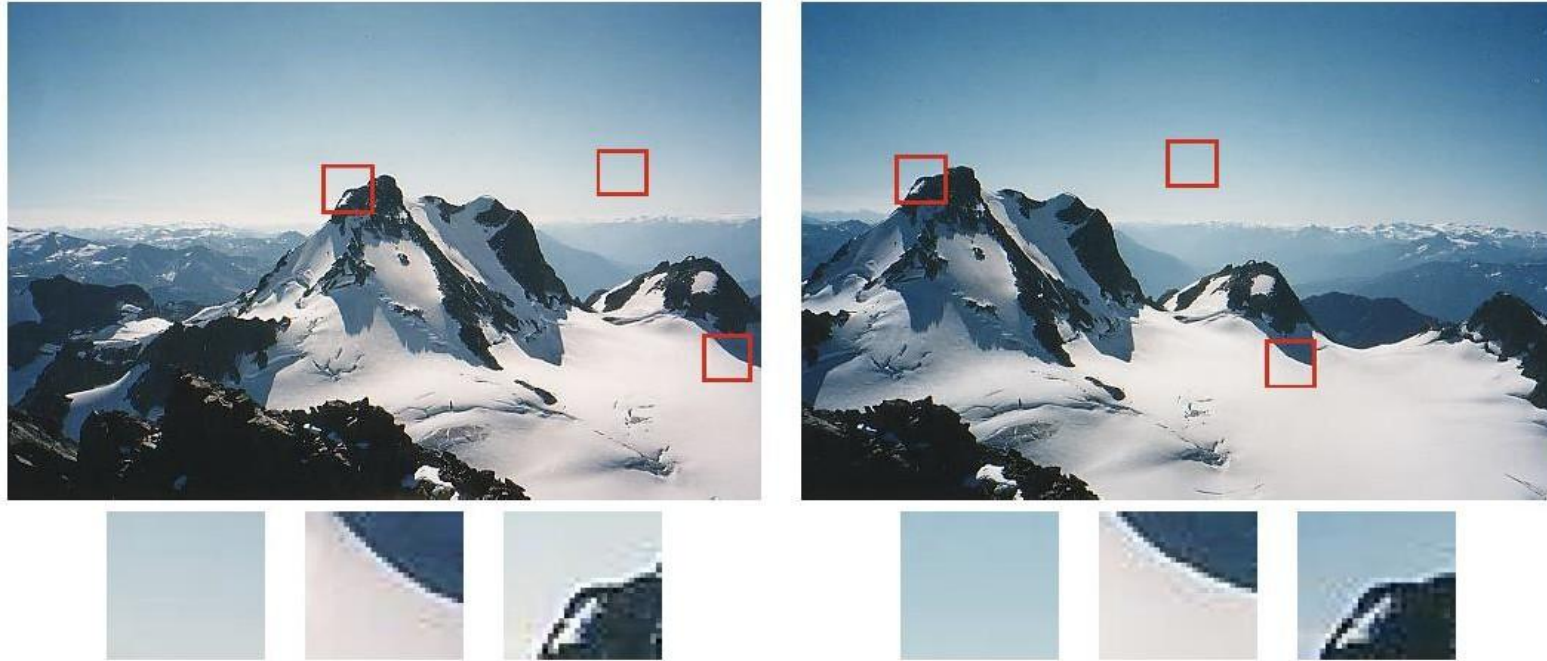


Applied Computer Vision

Features: Detection, Description, Matching, C
Tracking

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Recall



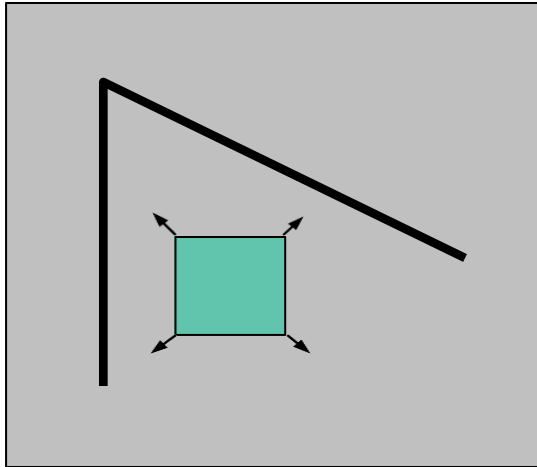
Flat surfaces are spread over a lot of area. It is difficult to find the exact location of these patches.

Edges are much more simple. You can find an approximate location, but exact location is still difficult.

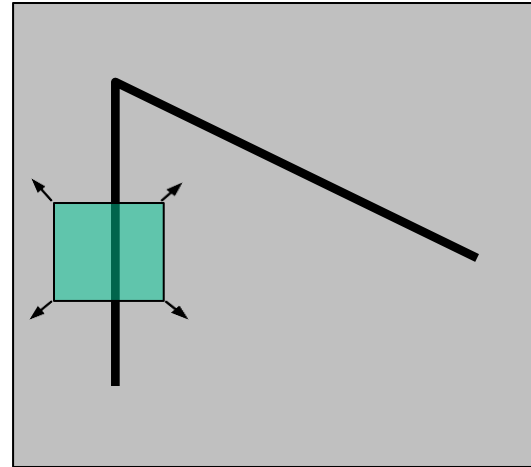
Corners can be “easily” found.

Finding interest points in an image

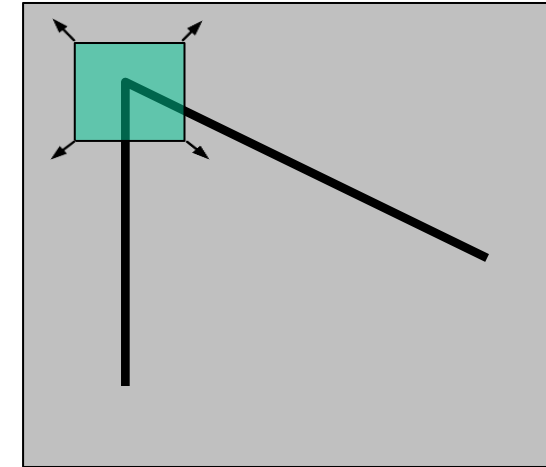
How does the window change when you shift it?



“flat” region: no change in all directions



“edge”:
no change along the edge direction



“corner”:
significant change in all directions, i.e., even the minimum change is large

Find locations such that the minimum **change** caused by shifting the window in any direction is large

Summary

- Feature Detection

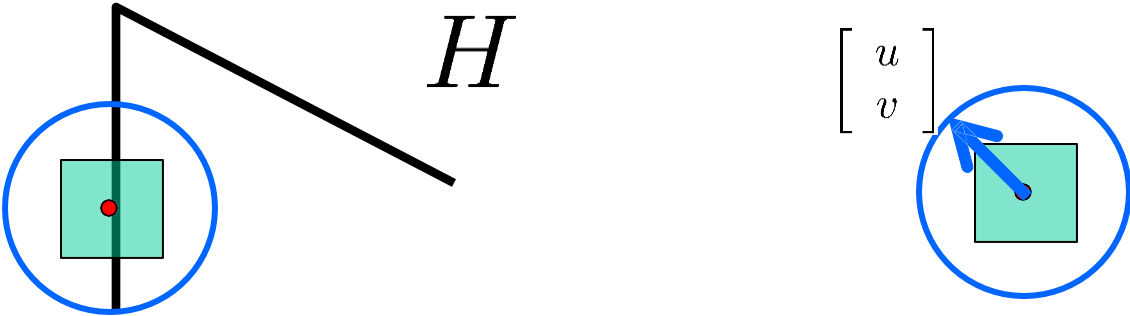
When performing feature detection, we do not know which other image locations the feature will end up being matched against.

Therefore, we can only **compute how stable this metric is with respect to small variations in position** (u, v) by **comparing an image patch against itself**, which is known as an **auto-correlation function or surface**

$$E(u, v) = \sum_{x, y} w(x, y) [I(x + u, y + v) - I(x, y)]^2$$

Feature detection: the math

This can be rewritten:

$$E(u, v) \approx [u \ v] \underbrace{\left(\sum_{(x,y) \in W} \begin{bmatrix} I_x^2 & I_x I_y \\ I_x I_y & I_y^2 \end{bmatrix} \right)}_H \begin{bmatrix} u \\ v \end{bmatrix}$$


For the example above:

- You can move the center of the green window to anywhere on the blue unit circle
- How do we find directions that will result in the largest and smallest E values?
- Find these directions by looking at the **eigenvectors** of **H**

Feature detection: the math

This can be rewritten:

$$E(u, v) \approx [u \ v] \underbrace{\left(\sum_{(x,y) \in W} \begin{bmatrix} I_x^2 & I_x I_y \\ I_x I_y & I_y^2 \end{bmatrix} \right)}_H \begin{bmatrix} u \\ v \end{bmatrix}$$

$$E(u, v) \approx [u \ v] \begin{bmatrix} S_{xx} & S_{xy} \\ S_{xy} & S_{yy} \end{bmatrix} \begin{bmatrix} u \\ v \end{bmatrix}$$

That matrix is the structure tensor:

$$H(x, y) = \begin{bmatrix} S_{xx} & S_{xy} \\ S_{xy} & S_{yy} \end{bmatrix}$$

Where:

$$S_{xx} = G_\sigma * I_x^2$$

$$S_{yy} = G_\sigma * I_y^2$$

$$S_{xy} = G_\sigma * (I_x I_y)$$

Quick eigenvalue/eigenvector review

The **eigenvectors** of a matrix **A** are the vectors **x** that satisfy:

$$Ax = \lambda x$$

The scalar λ is the **eigenvalue** corresponding to **x**

- The eigenvalues are found by solving:

$$\det(A - \lambda I) = 0$$

Feature detection: the math

Want $E(u,v)$ to be *large* in *all* directions

- the *minimum* of $E(u,v)$ should be large over all unit vectors $[u \ v]$
- this minimum is given by the smaller eigenvalue λ_2 of $\mathbf{H}$
- Look for large values of λ_2

Harris Corner Detector

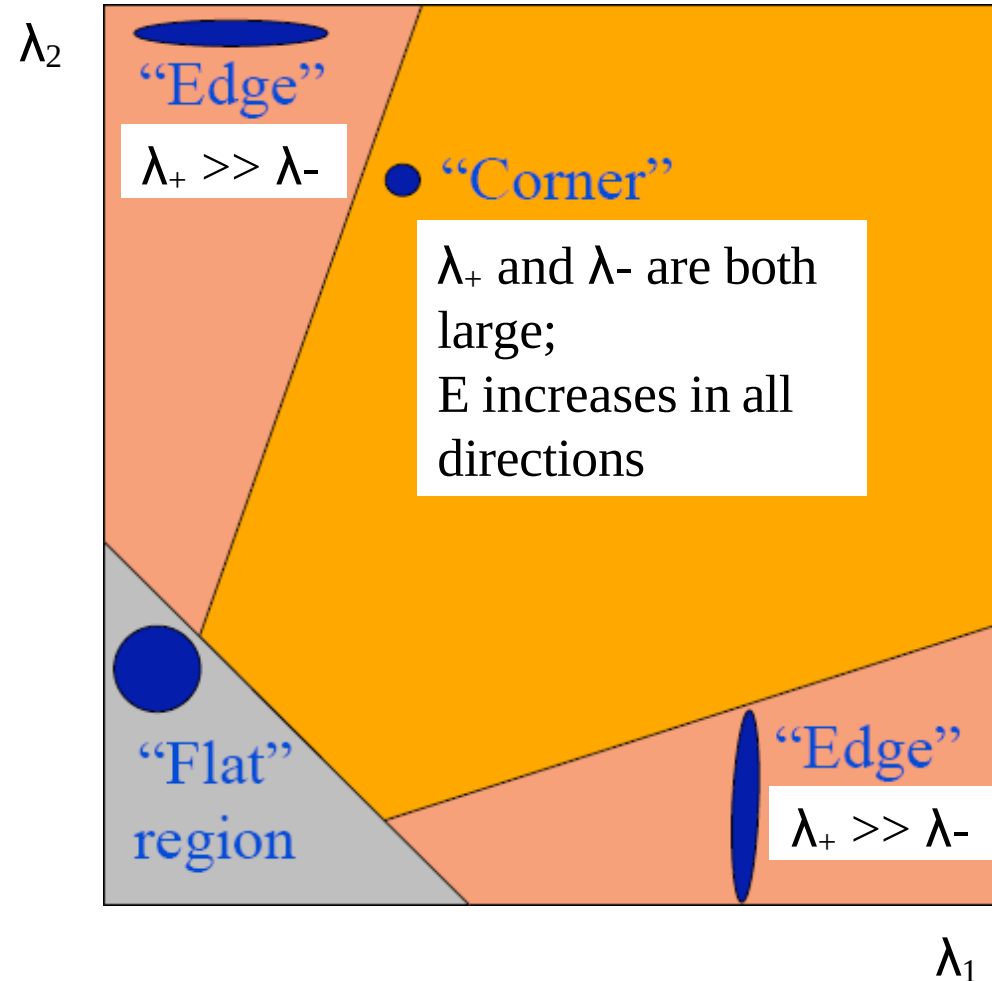
Scoring function

$$R = \det(H) - k * (\text{trace}(H))^2 = \lambda_+ \lambda_- - k * (\lambda_+ + \lambda_-)^2$$

- When $|R|$ is small, which happens when λ_+ and λ_- are small, the region is flat.
- When $R < 0$, which happens when $\lambda_+ \gg \lambda_-$, the region is edge.
- When R is large, which happens when λ_+ and λ_- are large and $\lambda_+ \sim \lambda_-$, the region is a corner.

Feature detection: the math

How are λ_+ , $\mathbf{x}_+$, λ_- , and $\mathbf{x}_-$ relevant for feature detection?



Feature detection (interest point detection) summary

Here's what you do

- Compute the gradient at each point in the image
- Create the ***H*** matrix from the entries in the gradient
- Compute the eigenvalues.
- Find points with large response ($\lambda_- > \text{threshold}$)
- Choose those points where λ_- is a local maximum as features (interest points)

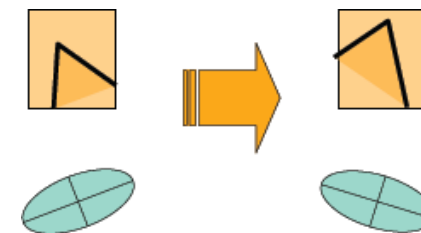
Shi-Tomasi Corner Detector

$$R = \min(\lambda_1, \lambda_2) = \lambda_-$$

Invariance of Eigenvalue-based feature detectors

Suppose you **rotate** the image by some angle

- Will you still pick up the same feature points?
- Yes (since eigenvalues remain the same)

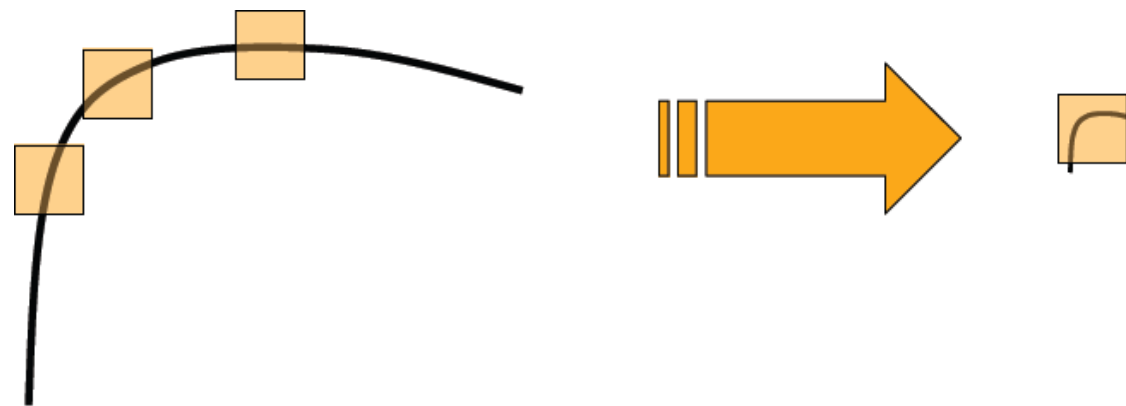


What if you change the brightness?

- Will you still pick up the same feature points?
- Mostly yes (uses gradients which involve pixel differences)

Scale?

- No!



All points will be
classified as **edges**

Corner !

Scale Invariance

In many applications, **the scale of the object of interest may vary in different images.**



Simple but inefficient solution:

Extract features at many different scales.

Match them to the object's known features at a particular scale.

More efficient solution:

Extract features that are invariant to scale.

Scale Invariance



Bird template

Scale Invariance



Bird template

Scale Inv

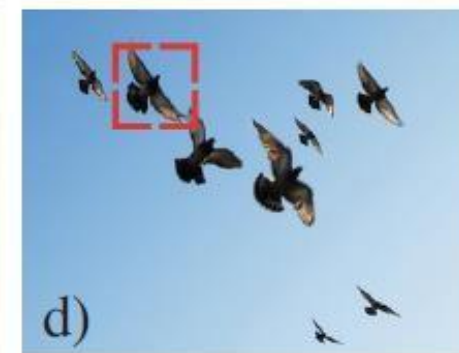
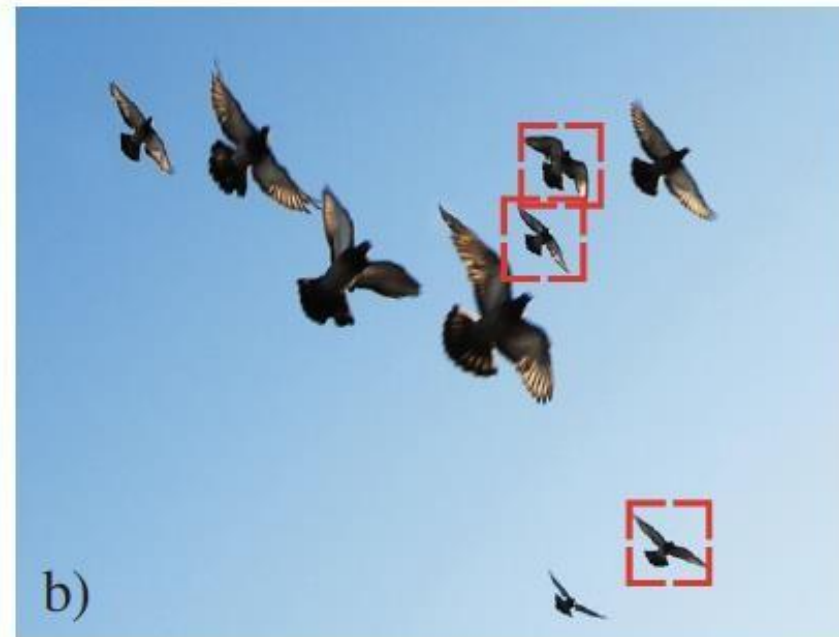


Image Pyramid

- Commonly used in coarse-to-fine matching, optical flow, stereo, blending, ...
- ... deep neural networks

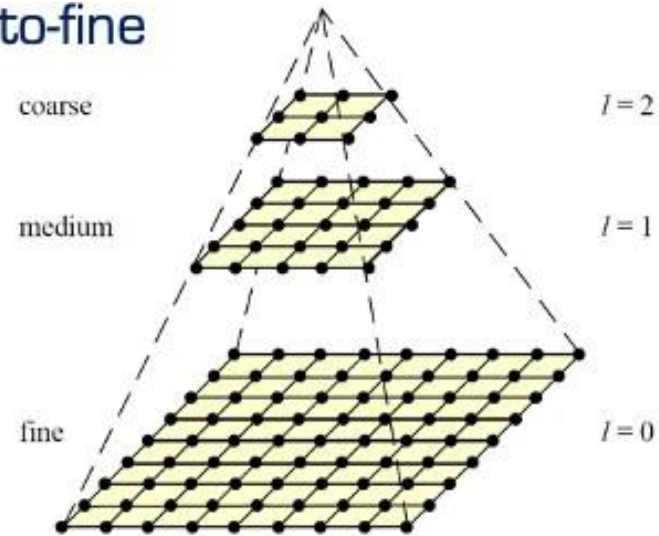


Figure 3.32 A traditional image pyramid: each level has half the resolution (width and height), and hence a quarter of the pixels, of its parent level.

Scale Invariance

Scale Invariant Feature Transform (SIFT) [Low04].

Convolve input image I with Gaussian G of various scale σ :

$$L(x, y, \sigma) = G(x, y, \sigma) * I(x, y) \quad (16)$$

where

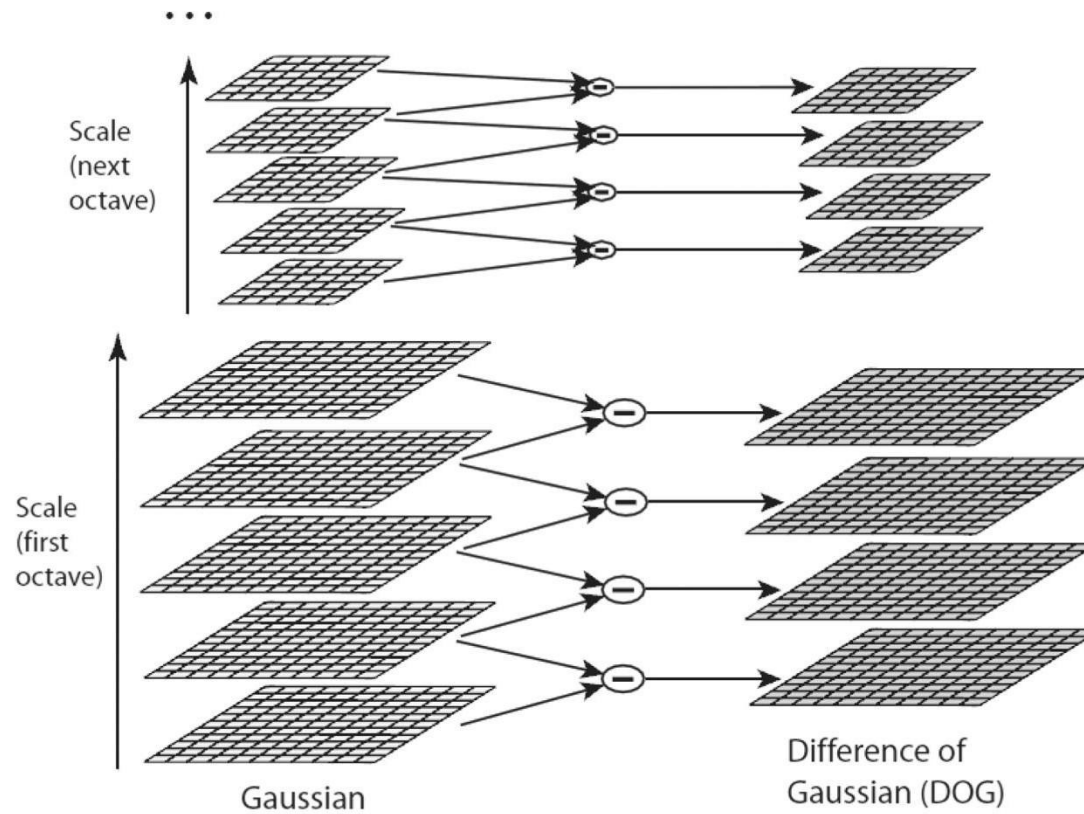
$$G(x, y, \sigma) = \frac{1}{2\pi\sigma^2} \exp\left(-\frac{x^2 + y^2}{2\sigma^2}\right). \quad (17)$$

This produces L at different scales.

To detect stable keypoint, convolve image I with difference of Gaussian:

$$\begin{aligned} D(x, y, \sigma) &= (G(x, y, k\sigma) - G(x, y, \sigma)) * I(x, y) \\ &= L(x, y, k\sigma) - L(x, y, \sigma). \end{aligned} \quad (18)$$

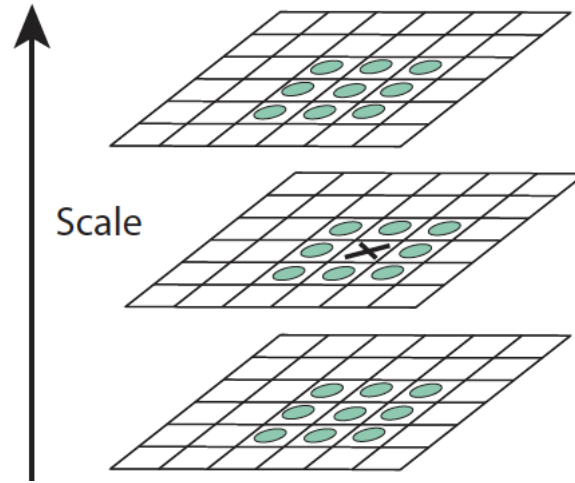
Scale Invariance



- Have 3 different scales within each octave (doubling of σ).
- Successive DOG images are subtracted to produce D .
- D images in a lower octave are downsampled by factor of 2.

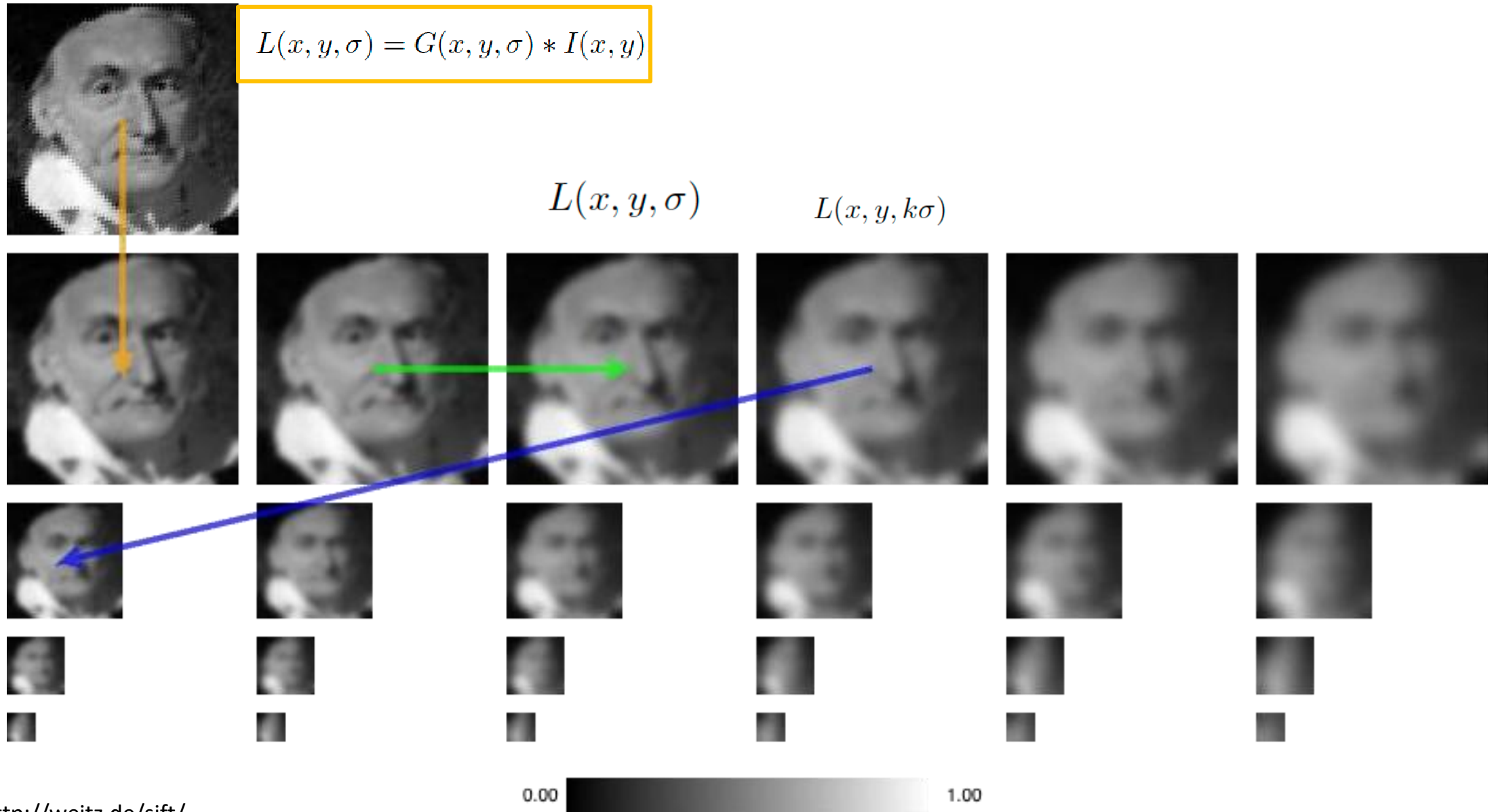
Difference of Gaussian

$$\begin{aligned} D(x, y, \sigma) &= (G(x, y, k\sigma) - G(x, y, \sigma)) * I(x, y) \\ &= L(x, y, k\sigma) - L(x, y, \sigma). \end{aligned}$$

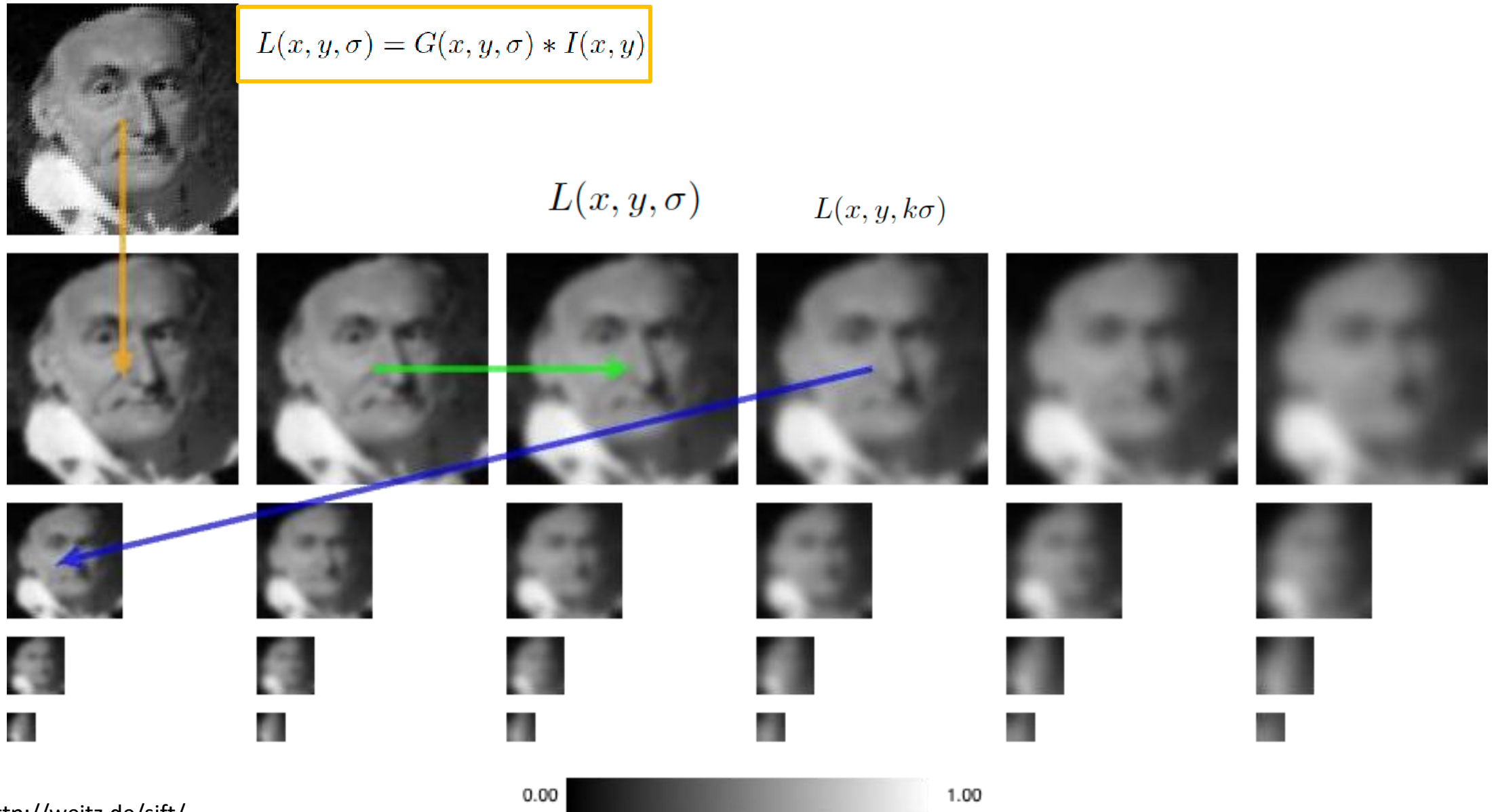


Maxima and minima of the difference-of-Gaussian images are detected by comparing a pixel (marked with X) to its 26 neighbors in 3x3 regions at the current and adjacent scales (marked with circles).

Scale-Space Representation

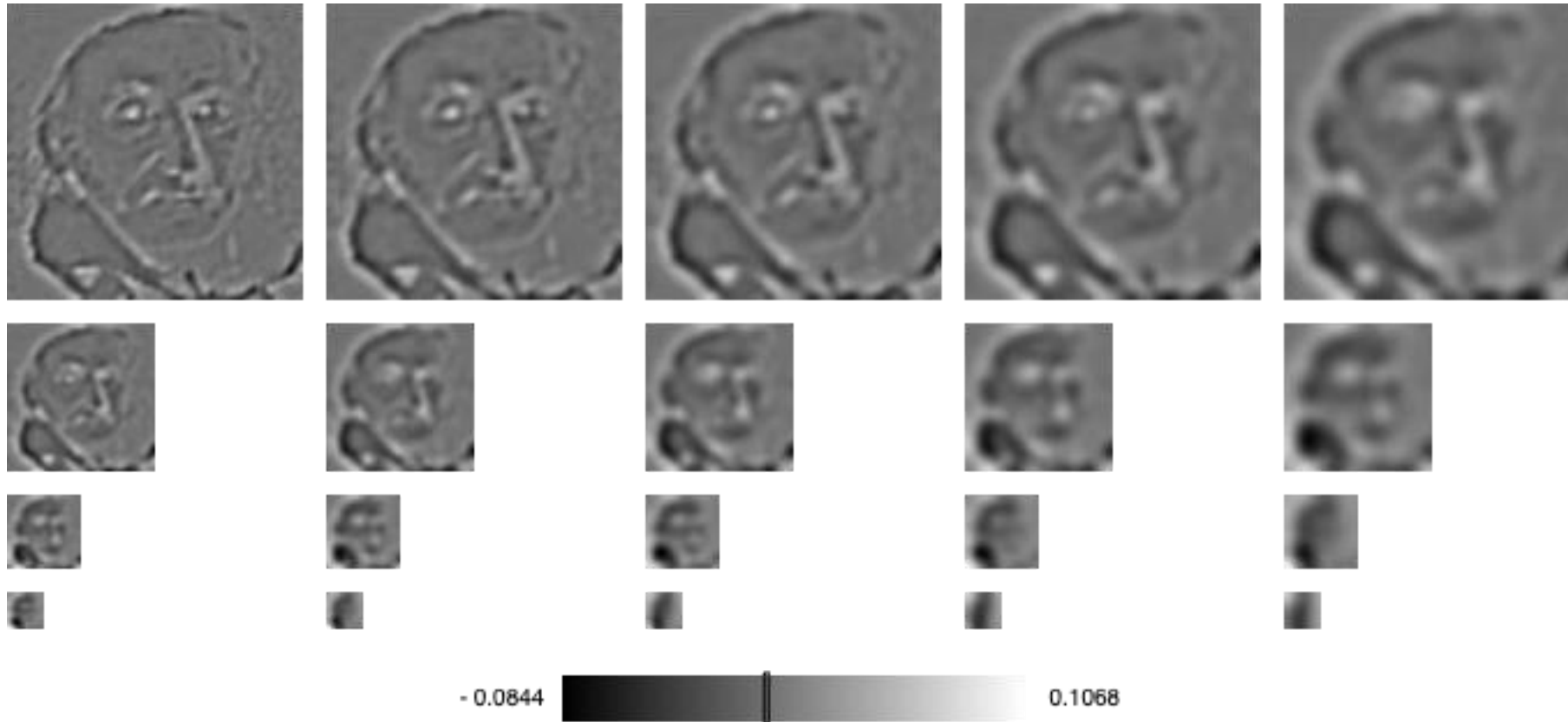


Scale-Space Representation



Difference of Gaussian

$$\begin{aligned} D(x, y, \sigma) &= (G(x, y, k\sigma) - G(x, y, \sigma)) * I(x, y) \\ &= L(x, y, k\sigma) - L(x, y, \sigma). \end{aligned}$$



Scale Invariant Features

